

ARYAN SRIVASTAVA

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PROFESSIONAL SUMMARY

B.Tech Electrical & Electronics Engineering student with an hands-on experience in embedded systems, IoT, automation, and data analytics. Experienced in developing hardware-software projects using Python, C/C++, MATLAB, ESP32, Arduino, SQL, Excel, and Power BI. Strong in problem-solving, project execution, team leadership, and stakeholder coordination, with experience leading large-scale social-impact initiatives.

TECHNICAL SKILLS

Programming: Python, C/C++, Embedded C/C++, SQL | **Engineering:** MATLAB, Simulink, PID Control, PWM, Motor Control
IoT & Embedded: Arduino, ESP32, IoT, Sensors, UART, I2C, Wi-Fi | **Tools:** Arduino IDE, Jupyter Notebook, Blynk IoT
Data & Analytics: Advanced SQL, Advanced Excel, Power BI, Data Analysis, Data Visualization
Areas: Embedded Systems, Automation, Control Systems, Energy Monitoring, Condition Monitoring

PROJECTS

Temperature-Based Variable-Speed BLDC Motor Controller

MATLAB | Simulink | Arduino Uno | Embedded C/C++ | PID | PWM

- Designed and built a closed-loop BLDC motor control prototype using temperature feedback; tuned PID parameters and applied software averaging to reduce oscillations and sensor noise.
- Developed the MATLAB/Simulink model, implemented control algorithms on Arduino Uno, and integrated LM35 temperature sensing, PWM generation, BLDC driver, and MOSFET switching.
- Compared PID-based variable-speed control with ON/OFF control; analyzed temperature response, PWM duty cycle, and power-consumption behavior.

IoT-Based Electrical Energy Monitoring System

ESP32 | PZEM-004T | Embedded C/C++ | Wi-Fi | UART | Blynk IoT

- Developed an IoT system for real-time monitoring of voltage, current, active power, power factor, and energy consumption.
- Interfaced PZEM-004T with ESP32 using UART, implemented Wi-Fi communication, and built a real-time IoT dashboard.
- Added over/under-voltage alerts and tested the system with multiple electrical loads for accurate energy analytics.

AC Energy Consumption & Cost Analysis Dashboard

Advanced Excel | Power BI | DAX | Data Analysis

- Developed an interactive dashboard for analyzing AC energy consumption, electricity cost, efficiency, savings, and payback periods.
- Built data models to analyze inverter vs. non-inverter operation, translating technical energy data into clear cost insights.

Gesture-Controlled Wireless Robotic Vehicle

ESP32 | Arduino IDE | Embedded C/C++ | MPU6050 | Motor Control

- Designed and built a wireless robotic vehicle controlled through hand gestures; implemented threshold logic to reduce unintended commands.

EXPERIENCE

Vice President (Dec 2025 – Present) & Camp Manager (Aug 2024 – Nov 2025) — BloodConnect Foundation

- Lead coordination and execution of large-scale blood-donation drives across multiple cities; coordinate regional leads and cross-functional teams.
- Manage stakeholder engagement with hospitals, institutions, and corporate partners to support operations and social-impact initiatives.
- Received *Rising Star – 2025* recognition for outstanding contribution, volunteer deployment, and donor engagement leadership.

Vice President — DIAS Club, Cambridge Institute of Technology

Nov 2025 – Present

- Lead communication workshops, debates, and public-speaking sessions to encourage structured student engagement.

Python Developer Intern — Motion Cut

Jul 2024 – Aug 2024

- Developed Python mini-applications; applied fundamentals including data types, loops, and structured problem-solving.

Industrial Level Internship (MATLAB) — Internshala

May 2025 – Jul 2025

- Applied numerical computation and data visualization for engineering applications. Earned Grade A and Industrial Training Certificate.

EDUCATION

B.Tech — Electrical & Electronics Engineering, Cambridge Institute of Technology, Ranchi **2023 – 2027** | CGPA: 8.00/10.00 (5th Sem)

Class XII, Kendriya Vidyalaya, Chandrapura | **Class X**, De-Nobili School, Chandrapura

XII: 68.6% | X: 79.4%

CERTIFICATIONS

E-Mobility with IoT (Reliance Foundation/Skill India) | **IoT Network Specialist** (Reliance Foundation/Skill India) | **Data Analytics Essentials** | **MATLAB** (Internshala/Skill India & NSDC) | **Complete Python Bootcamp: From Zero to Expert** – Udemy